

Justify all answers. Submit as a single PDF.

1. What time does a 12-hour clock read
 - (a) (5 points) 80 hours after it reads 11:00?
 - (b) (5 points) 40 hours before it reads 12:00?
 - (c) (5 points) 100 hours after it reads 6:00?
2. (10 points) Prove that if a is a positive integer, then 4 does not divide $a^2 + 2$.
3. (10 points) Prove that if n is an odd positive integer, then $n^2 \equiv 1 \pmod{8}$.
4. Use the extended Euclidean algorithm to express:
 - (a) (5 points) the $\gcd(26, 91)$ as a linear combination of 26 and 91.
 - (b) (5 points) the $\gcd(252, 356)$ as a linear combination of 252 and 356.
 - (c) (5 points) the $\gcd(144, 89)$ as a linear combination of 144 and 89.
 - (d) (5 points) the $\gcd(1001, 100001)$ as a linear combination of 1001 and 100001.
5. Note that among any three consecutive odd integers p , $p+2$, $p+4$, exactly one is divisible by 3.
 - (a) (5 points) Find three consecutive odd positive integers that are all prime; that is, find a triple of the form p , $p+2$, $p+4$ consisting entirely of primes.
 - (b) (5 points) Prove that, apart from the triple you found in part (a), there is no other triple of three consecutive odd positive integers that are all prime.
6. (10 points) Prove that the product of any three consecutive integers is divisible by 6.
7. Solve (find all solutions to) each of these congruences:
 - (a) (5 points) $34x \equiv 77 \pmod{89}$
 - (b) (5 points) $144x \equiv 4 \pmod{233}$
 - (c) (5 points) $200x \equiv 13 \pmod{1001}$
8. (10 points) Show with the help of Fermat's little theorem that if n is a positive integer, then 42 divides $n^7 - n$. *Hint. Show $n^7 - n$ is divisible by 2 and that $n^7 \equiv n \pmod{3}$ and $\pmod{7}$ by Fermat; conclude $42 \mid n^7 - n$.*
9. In this problem, you will combine Fermat's little theorem with the Chinese remainder theorem.
 - (a) (5 points) Compute $5^{2003} \pmod{7}$, $5^{2003} \pmod{11}$, and $5^{2003} \pmod{13}$ using Fermat's little theorem.
 - (b) (5 points) Using your answers from part (a) and the Chinese remainder theorem, find $5^{2003} \pmod{1001}$ (note that $1001 = 7 \cdot 11 \cdot 13$).

10. (10 points) Find the amount of zeros at the end of $100!$.
11. In this question we use the *generalized Chinese Remainder Theorem*. A system $x \equiv a_i \pmod{m_i}$ is solvable if and only if $a_i \equiv a_j \pmod{\gcd(m_i, m_j)}$ for every pair (i, j) . When solvable, the solution set is unique modulo $\text{lcm}(m_1, m_2, m_3)$. For each part below, first check the consistency conditions, then (if consistent) find the solution class modulo the lcm; otherwise, briefly explain why no solution exists.
- (a) (5 points) $x \equiv 5 \pmod{6}$, $x \equiv 3 \pmod{10}$, $x \equiv 8 \pmod{15}$.
- (b) (5 points) $x \equiv 7 \pmod{9}$, $x \equiv 4 \pmod{12}$, $x \equiv 16 \pmod{21}$.

Note. When a solution exists, it is unique modulo the lcm of the moduli; here, $\text{lcm}(6, 10, 15) = 30$ and $\text{lcm}(9, 12, 21) = 252$.